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Project Title:	Feasibility Study for a Space-Based Test of the Strong Equivalence Principle Using Lunar Laser Ranging		
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Murphy, Thomas Williams (University of California, San Diego)		
Grant/Contract No.:	NAG 3 2889		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obsacurec by Earth's gravity		
	APOLLO (the Apache Point Observatory Lunar Laser-ranging Operation) is a next-generation lunar laser ranging (LLR) experiment that will advance the state of LLR to millimeter precision in an effort to test Einstein's Strong Equivalence Principle. LLR involves measuring the round-trip travel time of laser pulses bounced off of the retroreflector arrays placed on the moon by Apollo astronauts. We anticipate reaching the limit of ground-based range precision with this experiment, and as such, we will explore the prospects of improved measurement accuracy via laser ranging from a space platform. Millimeter-level range precision in the earth-moon distance will enable the most stringent tests to date of several aspects of general relativity. Current range precision is around 2 centimeters, so that we anticipate at least an order-of-magnitude gain in sensitivity to potential violations of the equivalence principle, time rate of change of the gravitational constant, and other measures of significance in understanding the nature of gravity. The expected dramatic improvement in range precision is a consequence of the large-aperture 3.5 meter telescope we will use at the Apache		

Task Description:	Point Observatory in southern New Mexico. We believe that the ultimate precision of APOLLO will be limited by systematic effects associated with observing from the ground--both in terms of fluctuating ground level and atmospheric delay effects. A component of our objective is to evaluate the feasibility of eventually operating a lunar ranging platform from space, free of atmospheric and ground perturbations. Gravity is the hardest of the fundamental forces to understand, owing to its feebleness compared with the other forces of nature. Our current theory of gravity is embodied by general relativity, which is at its root incompatible with particle physics and quantum theory. All efforts to unify the fundamental forces of nature have failed, and it could be that our understanding of gravity is incomplete. This suspicion is fueled by the apparent acceleration of the expansion of the Universe, indicating that gravity may be rupulsive at cosmological scales, or that there may be new forces in the Universe as yet undiscovered. By probing the earth-moon orbit to millimeter precision, APOLLO may potentially shed light on this pressing issue in modern physics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The order-of-magnitude increase in ranging precision expected from APOLLO will give more precise information on crustal deformations of the earth and moon.
Task Progress:	Our work has two main thrusts: experimental and theoretical. The experimental work is concentrated on developing a new generation lunar laser-ranging (LLR) facility called APOLLO. Under the NASA grant NAG8-1756, the APOLLO lunar ranging project moved from conceptual status to a state of substantial physical implementation. At the close of the grant period, the majority of the optical, mechanical, and electronic design is complete. Specifically, we purchased a customized high-power, short-pulse laser and mounted it onto the Apache Point 3.5 meter telescope in a semi-permanent manner. We have fabricated three thermally insulated enclosures for the laser and other electronics equipment within the telescope dome. We have developed a timing system based on a GPS-disciplined oscillator and commercial time-to-digital converters (TDCs) that is capable of interval timing with an uncertainty of 15 ps. We have also developed a custom logic controller that synchronizes the multiplexed ranging, controlling the laser fire, the transmit/receive beam switch, the detector gating, the TDC calibration, tracks and latches event times, and coordinates the fiducial attenuation. We have worked extensively on characterization of the avalanche photodiode (APD) arrays using a fast-pulse diode laser. This culminated in the PhD thesis of Jana Strasburg at the University of Washington. The APD arrays were developed at Lincoln Labs and shared with APOLLO in exchange for our time-domain characterization of the devices. Through this effort we confirmed that the APDs are adequate for millimeter range determination, but also learned about their limitations (especially at long wavelengths), and about potential biases introduced by temporal walk (delay) associated with exact illumination position on the detector element. This crucial discovery has led to a system design modification that will eliminate the potential bias from this phenomenon. APD arrays of this sort would be a natural detector choice for space-based interplanetary laser ranging. The theoretical phase of our work is directed at extracting improved constraints on fundamental physics parameters from lunar laser-ranging data and on investigating new issues that can be addressed from such observations. Two new contributions to the understanding of relativistic gravity derived from lunar laser ranging results were developed in the last couple years. First, it was shown that LLR data could test Newton's third law --- "action equals reaction" --- for the gravitational force between different materials. If a body composed of different materials is considered in which the distribution of the materials are also non-symmetrical, if the gravitational force between the so-distributed materials does not fulfill Newton's third law, then the body will self accelerate. For Earth, the asymmetrically distributed material is the planet's ocean water which floats on the planet's mantle material in approximate isostatic equilibrium. Averaging over the day, the possible self-acceleration of Earth is along its polar axis, and the tilt of that axis puts part of that self acceleration in the orbital plane of the Earth-Moon system. Testing for a sidereal polarization of the lunar orbit with LLR puts an upper limit of a couple parts in 10^13 in any difference in the active to passive gravitational mass ratio for water versus average Earth material. General relativity predicts that a spinning body will not follow the gravitational geodesic trajectories of non-spinning bodies. The spin angular momentum of a body couples to or "feels" the gravitational gradient field of the external world. I have shown that LLR data detects this additional acceleration of a spinning body which in this case is the Earth-Moon system considered as a whole. The gravitational gradient of the Sun acts in the post-Newtonian order of gravitational theory primarily on the Moon due to that body's orbital angular momentum; the resulting polarization of that orbit is about an order of magnitude greater than the present experimental limits of LLR's recent fits of the accumulated data. In the last several years several surprising features of cosmological evolution have been detected. The expansion of the universe is apparently accelerating; and a variation in the fine structure "constant" was reported from observations of spectral lines of very distant galaxies; although this latter observation has now been called into doubt. The results of LLR have significant implications for such cosmological issues, so in recent papers and proceedings contributions I have been stressing the cosmological implications of LLR results. LLR presently provides the best measurement of time variation of Newton's G; no variation is seen at better than a part in 10^12 per year level which is about a factor 80 slower than the Hubble expansion rate. The measured equality of the acceleration rate of Earth and Moon by the Sun (to 1.3 parts in 10^13) can be directly converted into a constraint on any spatial gradient of Newton's G produced by the matter of the Sun. Expressed in appropriate units, such a gradient of G is less than 2 10^-4 the gravitational acceleration produced by the Sun's matter. This reveals that LLR's test for a spatial gradient of G is under most circumstances a deeper probe of gravitational theory than is the time gradient of G measurement by a couple orders of magnitude. Only if the hypothesized dark matter or dark energy of the universe were a stronger source of space-time variation of G would the temporal gradient measurement detect violations of theory before LLR's spatial gradient tests. Because Earth and Moon are composed of different materials (Earth has a large iron-like core), equality of free fall rate of Earth and Moon toward the Sun also places strong constraint on any spatial gradient of the fine structure constant due to proximity to the Sun's matter. Converting this into the appropriate units, this spatial gradient test for the fine structure constant appears many orders of magnitude stronger than the best constraints on time variation of this parameter in physical theory.

	LLR can be used to test for fixed cosmic spatial gradients in G and/or the fine structure constant; if such gradients exist the lunar orbit will be polarized in a fixed sidereal direction. Because such a polarization of the orbit is so very close to being in resonance with the natural eccentric perturbation of the orbit (the frequency difference is 2pi/(9 years)), this test is extremely sensitive. We can conclude, for example, that the fine structure constant probably can not vary across our galaxy by more than a part in 10^10, and can not vary across a space-slice of the entire universe by more than a part in 10^4. Such experimental constraints from LLR can perhaps be useful to various cosmological models which include scalar fields which modify gravitation and constancy of fundamental parameters of physical theory. Eric Adelberger, Thomas Murphy, Jr. and Kenneth Nordtvedt
Bibliography Type:	Description: (Last Updated: 03/17/2011)
Articles in Peer-reviewed Journals	Williams JG, Turyshev SG, Murphy TW Jr. "Improving LLR tests of gravitational theory." International Journal of Modern Physics D. 2004 Mar;13(3): 567-82. http://dx.doi.org/10.1142/S0218271804004682 , Mar-2004
Significant Media Coverage	Davidson K. "PI's work described in article, 'Lasers shoot for the moon to test Einstein theory; precise measures of distance could change notions of gravity.'" San Francisco Chronicle, January 17, 2002. http://articles.sfgate.com/2002-01-17/news/17528520_1_einstein-s-theory-moon-precision , Jan-2003
Significant Media Coverage	Broad WJ. "Article describes PI's work, 'Moon's dust hides a throbbing heart.'" New York Times, Science section, May 14, 2002. http://www.nytimes.com/2002/05/14/science/moon-s-dust-hides-a-throbbing-heart.html?src=pm , Jan-2003
Significant Media Coverage	Knight W. "Article describes PI's work, ' Moon measurements will test gravity's principles.'" New Scientist, January 15, 2002. http://www.newscientist.com/article/dn1792-moon-measurements-will-test-gravitys-principles.html , Jan-2003